

1. Correlation Analysis of Student Performance

Import student datasets containing attendance, assignment marks, internal marks, study hours, and final marks. Explore the tables, join relevant datasets, calculate correlations between variables, identify outliers, and use appropriate charts to explain the major patterns in student performance.

SOLUTION:

Objective

To analyze the relationship between attendance, assignment marks, internal marks, study hours, and final marks and identify important patterns and outliers in student performance.

Dataset

Assume four CSV files:

- **student.csv → Student ID, Name, Department**
- **attendance.csv → Student ID, Attendance**
- **assignment.csv → Student ID, Assignment Marks**
- **marks.csv → Student ID, Internal Marks, Study Hours, Final Marks**

Complete Python Code

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

student = pd.read_csv("student.csv")
attendance = pd.read_csv("attendance.csv")
assignment = pd.read_csv("assignment.csv")
marks = pd.read_csv("marks.csv")

print("Student Table:")
print(student.head())
```

```

print("\nAttendance Table:")
print(attendance.head())
print("\nAssignment Table:")
print(assignment.head())
print("\nMarks Table:")
print(marks.head())
print("\nDataset Information:")
print(marks.info())
print("\nMissing Values:")
print(student.isnull().sum())
print(attendance.isnull().sum())
print(assignment.isnull().sum())
print(marks.isnull().sum())
df = student.merge(attendance, on="Student_ID")
df = df.merge(assignment, on="Student_ID")
df = df.merge(marks, on="Student_ID")
print("\nCombined Dataset:")
print(df.head())
print("\nStatistical Summary:")
print(df.describe())
correlation = df[
    ["Attendance", "Assignment_Marks",
     "Internal_Marks", "Study_Hours", "Final_Marks"]
].corr()
print("\nCorrelation Matrix:")
print(correlation)
print("\nCorrelation with Final Marks:")
print(correlation["Final_Marks"].sort_values(ascending=False))
Q1 = df["Final_Marks"].quantile(0.25)
Q3 = df["Final_Marks"].quantile(0.75)
IQR = Q3 - Q1

```

$\text{lower} = Q1 - 1.5 * IQR$

$\text{upper} = Q3 + 1.5 * IQR$

```
outliers = df[
    (df["Final_Marks"] < lower) |
    (df["Final_Marks"] > upper)
]
print("\nFinal Marks Outliers:")
print(outliers[["Student_ID", "Final_Marks"]])

plt.figure(figsize=(8, 6))
sns.heatmap(correlation, annot=True, cmap="coolwarm")
plt.title("Correlation of Student Performance Variables")
plt.show()

plt.figure(figsize=(7, 5))
sns.scatterplot(
    data=df,
    x="Study_Hours",
    y="Final_Marks"
)
plt.title("Study Hours vs Final Marks")
plt.xlabel("Study Hours")
plt.ylabel("Final Marks")
plt.show()

plt.figure(figsize=(7, 5))
sns.scatterplot(
    data=df,
    x="Attendance",
    y="Final_Marks"
)
plt.title("Attendance vs Final Marks")
plt.xlabel("Attendance (%)")
plt.ylabel("Final Marks")
plt.show()
```

```
# Final marks distribution and outliers

plt.figure(figsize=(7, 5))

sns.boxplot(y=df["Final_Marks"])

plt.title("Final Marks Outlier Detection")

plt.ylabel("Final Marks")

plt.show()
```

Sample Output

The exact numerical output depends on the student dataset used.

Combined Dataset

	Student_ID	Name	Department	Attendance	Assignment_Marks
0	101	Arun	CSE	92	85
1	102	Ravi	CSE	78	72
2	103	Kumar	ECE	95	90
3	104	Priya	CSE	65	60
4	105	Divya	ECE	88	82

After joining:

	Student_ID	Attendance	Assignment_Marks	Internal_Marks	Study_Hours	Final_Marks
0	101	92	85	88	6.5	91
1	102	78	72	74	4.0	76
2	103	95	90	92	8.0	94
3	104	65	60	62	2.5	65
4	105	88	82	85	6.0	87

Correlation Matrix

	Attendance	Assignment	Internal	Study_Hours	Final_Marks
Attendance	1.00	0.68	0.65	0.61	0.70
Assignment	0.68	1.00	0.79	0.72	0.82
Internal	0.65	0.79	1.00	0.74	0.88
Study_Hours	0.61	0.72	0.74	1.00	0.80
Final_Marks	0.70	0.82	0.88	0.80	1.00

Correlation with Final Marks

Internal_Marks	0.88
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Assignment_Marks 0.82

Study_Hours 0.80

Attendance 0.70

Final_Marks 1.00

Outlier Output

Final Marks Outliers:

Student_ID Final_Marks

118 35

145 98

Major Patterns

1. Internal marks have the strongest correlation with final marks (0.88), showing that students performing well in internal assessments generally perform well in final examinations.
2. Assignment marks also have a strong positive correlation (0.82) with final marks.
3. Study hours show a strong positive relationship (0.80) with final marks, indicating that students who spend more time studying tend to achieve better results.
4. Attendance has a positive correlation (0.70) with final marks, suggesting that regular attendance is associated with better academic performance.
5. Outliers represent unusually low or high final marks and should be investigated individually to determine whether they are genuine performance differences or data errors.

Conclusion

The analysis shows that internal marks, assignment performance, study hours, and attendance are positively associated with final marks. Internal marks have the strongest relationship with final performance in this example. The charts and correlation analysis help teachers identify students who may need additional academic support and mentoring.

2. Sales and Customer Pattern Analysis

Import sales and customer datasets and combine them using suitable table-joining techniques. Analyze the correlation between customer age, purchase frequency, discount, and purchase amount. Create meaningful customer groups, identify unusual purchasing patterns, and present the results using suitable visualizations.

SOLUTION:

Objective

To combine sales and customer datasets, study the relationship between customer age, purchase frequency, discount, and purchase amount, create meaningful customer groups, and identify unusual purchasing patterns.

Dataset

Assume two CSV files:

- customers.csv → Customer ID, Name, Age, Gender, Location
- sales.csv → Customer ID, Purchase Frequency, Discount, Purchase Amount, Product

Complete Python Code

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

customers = pd.read_csv("customers.csv")
sales = pd.read_csv("sales.csv")

print("Customer Dataset:")
print(customers.head())
print("\nSales Dataset:")
print(sales.head())
print("\nCustomer Dataset Information:")
print(customers.info())
print("\nSales Dataset Information:")
print(sales.info())
print("\nMissing Values in Customer Dataset:")
print(customers.isnull().sum())
print("\nMissing Values in Sales Dataset:")
print(sales.isnull().sum())
```

```

df = pd.merge(
    customers,
    sales,
    on="Customer_ID",
    how="inner"
)

print("\nCombined Dataset:")
print(df.head())
print("\nStatistical Summary:")
print(df.describe())
correlation = df[
    ["Age", "Purchase_Frequency",
     "Discount", "Purchase_Amount"]
].corr()
print("\nCorrelation Matrix:")
print(correlation)
print("\nCorrelation with Purchase Amount:")
print(
    correlation["Purchase_Amount"]
    .sort_values(ascending=False)
)

def customer_group(row):
    if row["Purchase_Frequency"] >= 10 and row["Purchase_Amount"] >= 5000:
        return "High Value"
    elif row["Purchase_Frequency"] >= 5:
        return "Regular"
    else:
        return "Occasional"

df["Customer_Group"] = df.apply(
    customer_group,
    axis=1
)

```

```

print("\nCustomer Groups:")
print(df["Customer_Group"].value_counts())

group_analysis = df.groupby(
    "Customer_Group"
)[
    ["Age", "Purchase_Frequency",
     "Discount", "Purchase_Amount"]
].mean()

print("\nCustomer Group Analysis:")
print(group_analysis)

Q1 = df["Purchase_Amount"].quantile(0.25)
Q3 = df["Purchase_Amount"].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
unusual = df[
    (df["Purchase_Amount"] < lower) |
    (df["Purchase_Amount"] > upper)
]

print("\nUnusual Purchasing Patterns:")
print(
    unusual[
        ["Customer_ID", "Age",
         "Purchase_Frequency",
         "Discount", "Purchase_Amount"]
    ]
)

plt.figure(figsize=(7, 5))
sns.scatterplot(
    data=df,
    x="Age",
    y="Purchase_Amount"
)

```

```
plt.title("Customer Age vs Purchase Amount")
plt.xlabel("Customer Age")
plt.ylabel("Purchase Amount")
plt.show()
plt.figure(figsize=(7, 5))
sns.scatterplot(
    data=df,
    x="Purchase_Frequency",
    y="Purchase_Amount"
)
plt.title("Purchase Frequency vs Purchase Amount")
plt.xlabel("Purchase Frequency")
plt.ylabel("Purchase Amount")
plt.show()
plt.figure(figsize=(7, 5))
sns.countplot(
    data=df,
    x="Customer_Group"
)
plt.title("Customer Group Distribution")
plt.xlabel("Customer Group")
plt.ylabel("Number of Customers")
plt.show()
plt.figure(figsize=(7, 5))
sns.boxplot(
    data=df,
    x="Customer_Group",
    y="Purchase_Amount"
)
plt.title("Purchase Amount by Customer Group")
plt.xlabel("Customer Group")
plt.ylabel("Purchase Amount")
```

plt.show()

Sample Output

The exact values will depend on the dataset used.

Combined Dataset

	Customer_ID	Name	Age	Gender	Location
0	101	Arun	22	Male	Chennai
1	102	Ravi	35	Male	Mumbai
2	103	Priya	28	Female	Chennai
3	104	Divya	42	Female	Delhi
4	105	Karthik	31	Male	Bangalore

	Purchase_Frequency	Discount	Purchase_Amount
0	12	10	6500
1	5	15	2800
2	8	10	4200
3	3	20	1800
4	7	5	3500

Correlation Matrix

	Age	Purchase_Frequency	Discount	Purchase_Amount
Age	1.00	0.15	-0.05	0.22
Purchase_Frequency	0.15	1.00	0.12	0.78
Discount	-0.05	0.12	1.00	-0.18
Purchase_Amount	0.22	0.78	-0.18	1.00

Interpretation

- Purchase Frequency and Purchase Amount = 0.78 → strong positive relationship.
- Age and Purchase Amount = 0.22 → weak positive relationship.
- Discount and Purchase Amount = -0.18 → weak negative relationship.

Drag points

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Linear correlation is positive ($r = 0.98$).

NegativeNear zeroPositive

NegativeNear zeroPositive

Give feedback

Customer Groups

Customer Groups:

Regular 45

Occasional 35

High Value 20

Group Analysis

	Age	Purchase_Frequency	Discount	Purchase_Amount
High Value	34.5	13.2	8.4	6850
Regular	31.8	7.1	11.2	3750
Occasional	29.6	2.8	15.5	1650

Unusual Purchasing Patterns

Customer_ID	Age	Purchase_Frequency	Discount	Purchase_Amount
118	45	15	5	12500
145	23	1	0	250

These customers have unusually high or low purchase amounts and should be investigated.

Major Findings

1. Purchase frequency has the strongest relationship with purchase amount, meaning frequent customers generally spend more.
2. High-value customers purchase frequently and have higher total spending, making them important customers for the business.
3. Age has only a weak relationship with purchase amount, so age alone is not a strong predictor of customer spending.
4. Discount has a weak negative relationship with purchase amount in the sample, meaning larger discounts do not necessarily result in larger purchases.

5. Unusual purchasing patterns can identify potential bulk buyers, inactive customers, or suspicious transactions.

Recommendations

- Provide loyalty rewards to high-frequency customers.
- Use personalized offers for regular customers.
- Investigate unusually high-value purchases for potential bulk sales or unusual transactions.
- Avoid relying only on age for customer targeting.
- Analyze discount strategies to determine which offers actually increase purchase frequency and revenue.

Conclusion

The combined sales and customer analysis shows that purchase frequency is an important factor associated with purchase amount. Customer segmentation helps the company identify high-value, regular, and occasional customers, while outlier detection helps identify unusual purchasing behavior. These insights can improve customer targeting, loyalty programs, discount strategies, and sales planning.